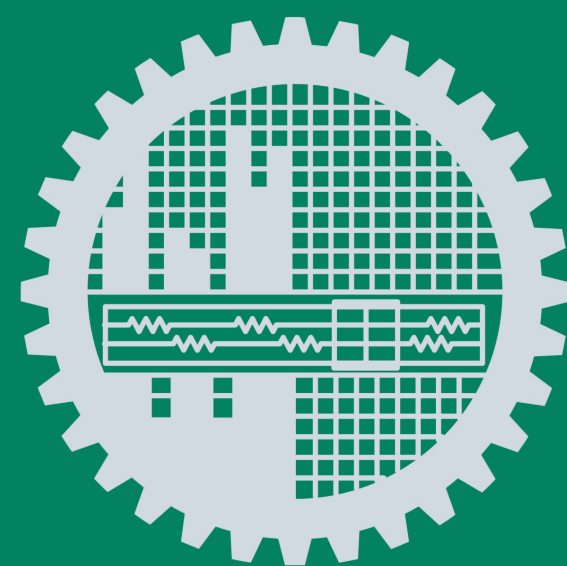




Agentic Graph Reasoning: Autonomous Knowledge Graph Navigation for Fact Verification and Hallucination Mitigation in Large Language Models

Md. Sakif Khan (0421052099) ▪ Supervisor: Dr. Sadia Sharmin, Associate Professor
Department of Computer Science and Engineering, BUET

Problem

Large language models answer factual questions fluently — and sound just as fluent when they do not hold the fact.

Retrieval-Augmented Generation retrieves *once*, before any reasoning starts, so a multi-hop question must be answered from whatever that first query returned. Graph-navigating systems do better by interleaving retrieval with reasoning.

But all of them emit whatever the final generation call produces. **Nothing checks what the answer asserts before the user sees it.**

The gap, measured

A parametric LLM control, on the same questions, asserted entities that *do not exist anywhere in the graph*:

22.1 %

221 of 1,001 asserted entities ungrounded — and stated without a hedge

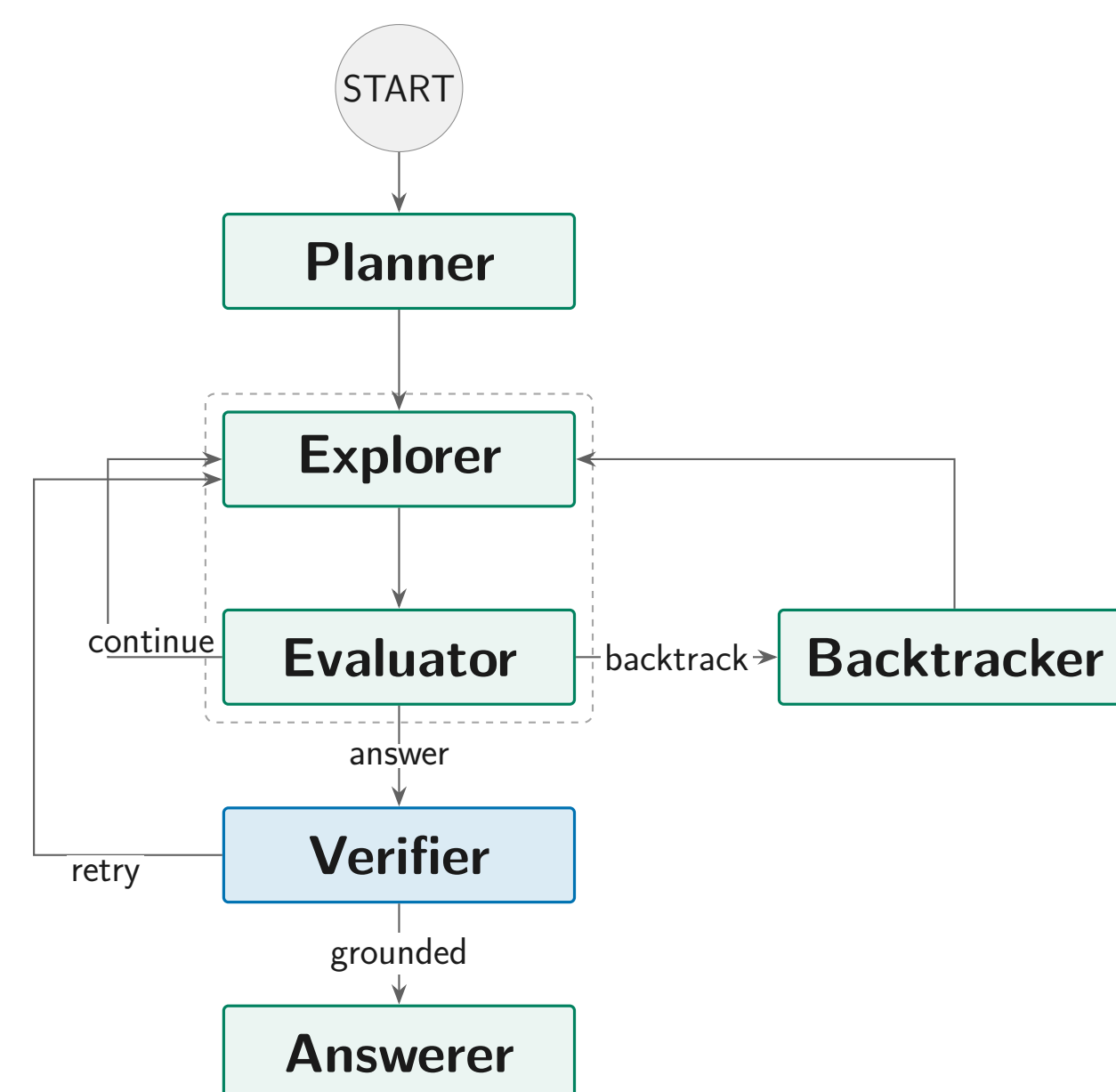
Retrieval reduces this without closing it. Only a system that checks its own claims against what it actually traversed can.

Research questions

- RQ1** Does agentic navigation improve multi-hop accuracy — and does the advantage *grow* with hop count?
- RQ2** What does pre-generation verification contribute beyond navigation alone?
- RQ3** Which components earn their cost, in accuracy and in tokens?

Method: AGR

An **explicit state machine**, not a prompt loop. Six nodes; the model is consulted only at decision points, and every cycle passes a router checking a monotone counter — so termination never depends on the model cooperating.



The dashed box is the search loop: a **guided beam** of width 3 over candidate edges, scored by embedding similarity blended with a model judgement of the active sub-objective.

Contribution: Structural Verification

Before the answer is emitted, the draft is split into **atomic claims** and each is checked against *the triples the agent actually traversed* — not against the model's own belief.

- Grounded claims return **paired with their supporting triples**.
- Claims that cannot be grounded are re-explored, then withdrawn.
- The system therefore **hedges rather than asserts** when the graph does not support it.

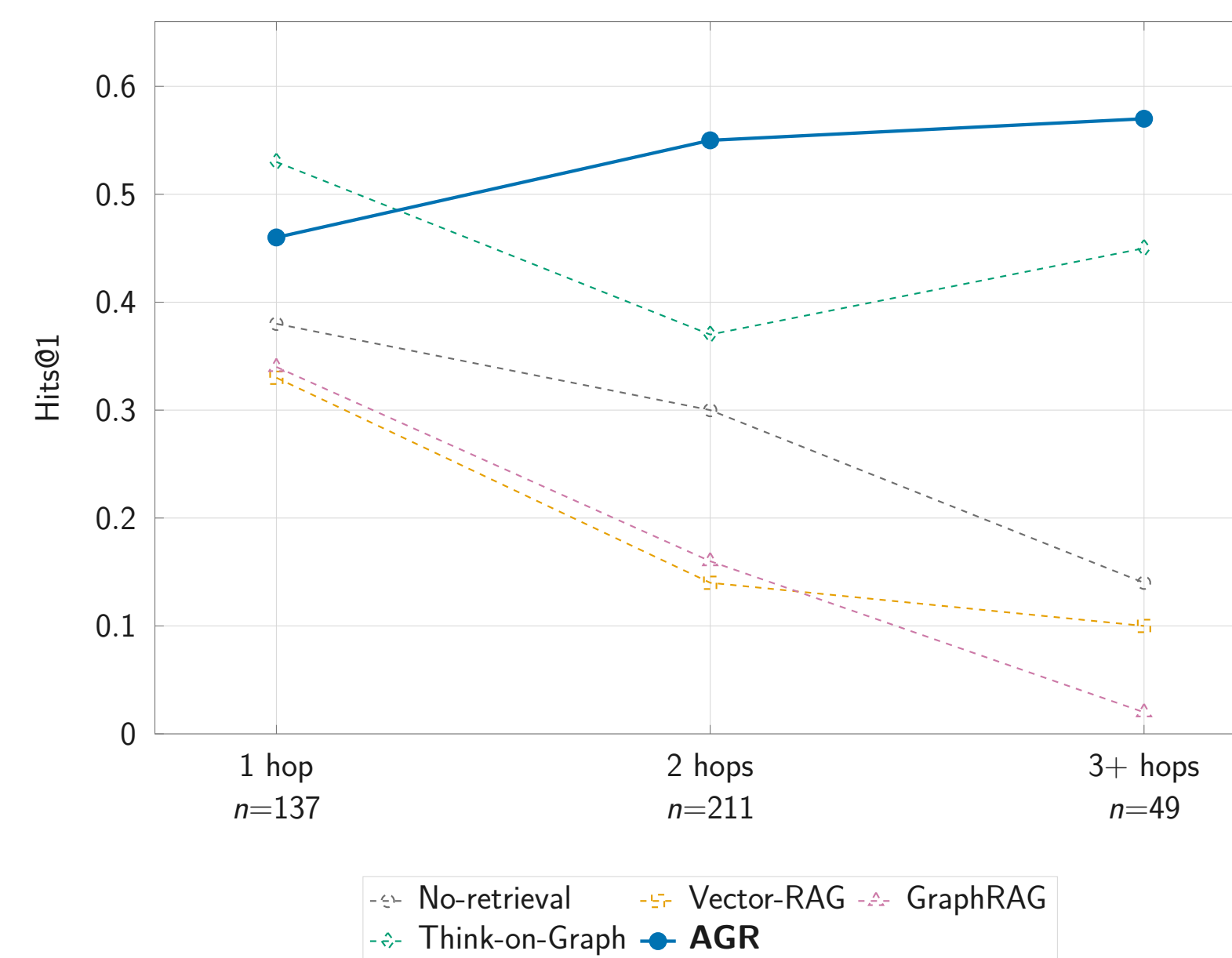
Results

Five systems, **one frozen backbone**, one shared call budget, one environment of 2.59M entities and 8.31M triples — so any difference is architecture, not model capacity. 400 certified questions per dataset.

System	WebQSP		CWQ	
	H01	F1	H01	F1
No-retrieval	0.453	0.305	0.307	0.279
Vector-RAG	0.568	0.432	0.203	0.168
Static GraphRAG	0.340	0.262	0.205	0.183
Think-on-Graph	0.660	0.477	0.435	0.378
AGR (ours)	0.755	0.642	0.522	0.469

Ahead of every baseline on both datasets, at 6.2 and 8.9 model calls against Think-on-Graph's 12.8 and 18.2 — about half the calls.

RQ1: the advantage grows with hops



AGR alone ends above where it started. Every other system decays as the questions deepen; AGR rises from 0.46 to 0.57.

Key findings

0 / 1,709

asserted entities ungrounded, both datasets pooled — AGR never names an entity absent from the graph

29 % / 44 %

of questions on which the agentic baseline exhausts the shared call budget. AGR's call budget *never* binds

An honest caveat. On the questions Think-on-Graph does finish, it leads AGR. The whole aggregate margin comes from the ones it cannot afford to finish — which is the finding, not a footnote.

Verification does not raise accuracy. It converts silent error into an explicit hedge and attaches evidence. Its contribution is auditability, and this poster says so because that is what was measured.

Conclusion

- Interleaving retrieval with reasoning, under *explicit budgets*, beats both one-shot retrieval and unbounded beam search.
- Checking claims against the traversed subgraph before emitting gives an answer that arrives **with its evidence**.
- Everything here is **training-free**: one frozen backbone, no fine-tuning, no gradient step.
- Every reported number is regenerated from committed run records, and every question identifier is published.

Acknowledgements

I thank my supervisor, Dr. Sadia Sharmin, and the Department of CSE, BUET. The environment is built from the public Reasoning-on-Graphs subgraph release over WebQSP and CWQ.